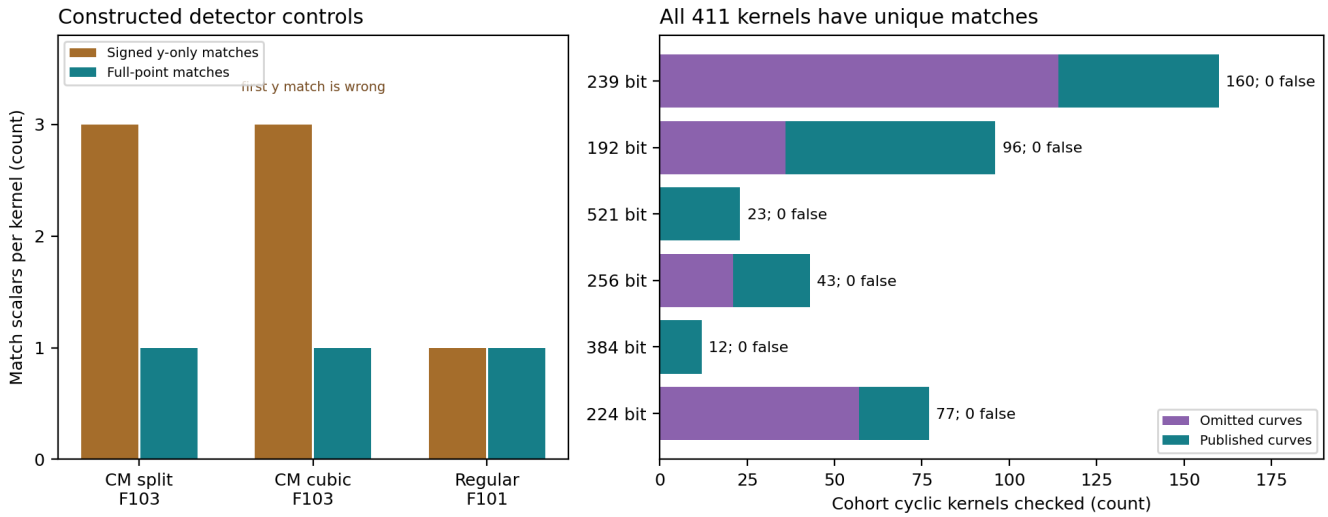
CM controls and Frobenius matching

2026-10-27 | T-5518 | ATKIN-Y-MATCH-01

Observations only. The reconstructed y-only early-stop failure occurs on a CM positive control. It occurs on **none of the 411 retained cohort kernels**. Every one has nonzero S1 and a unique coincident signed y-only/full-point match. The historical selector remains unidentified.

Two independent implementations agree on 415 cases each: 411 cohort kernels plus four control calls. The verifier computes 29,401 positive-half scalar-coordinate pairs using division-polynomial derivative jets. Each index is compared with both signs, giving 58,802 signed comparisons per implementation. No timing comparison is claimed.

Field / stream	Omitted	Published	Kernels	Ambiguous
239 / 863711343	5	3	160	0
192 / 863966734	2	3	96	0
521 / 871002724	0	1	23	0
256 / 871696354	1	1	43	0
384 / 872952608	0	1	12	0
224 / 891607927	3	1	77	0



Exact counts on the fixed finite domain: 228 kernels on the 11 omissions; 183 on the ten publications. Kernels are nested within curves; no independent-sample or population inference is made.

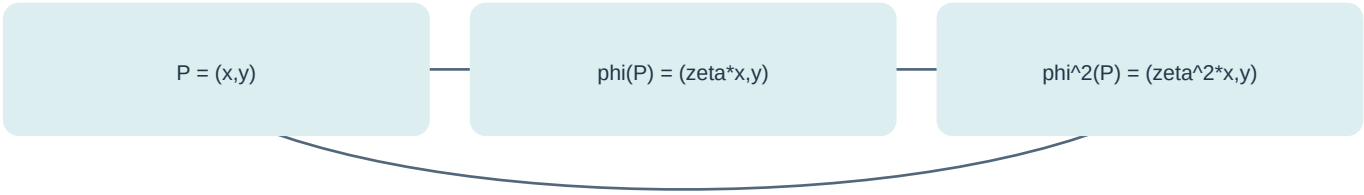
Working CM detector. For E/F103: y^2=x^3+1, the irreducible 7-kernel has y-only matches {3,5,6}, full match {5}, and first encountered match 6. Stopping there implies trace 1 modulo 7 instead of the correct 6. The split 7-kernel also has three y-only matches. The regular F101 control and its A/A each have one correct match.

A kernel condition behind the ambiguity

Derived elementary argument; no novelty or historical-use claim. Let H be a Frobenius-stable cyclic subgroup of odd prime order ell, with Frobenius scalar lambda. Let K be its monic half-kernel polynomial and S1 its root sum.

If a different scalar k has the same complete y-function, a=k/lambda preserves y on every nonidentity point. Its orbits fit in fibers of a cubic, so their size is at most three. Order two is impossible: a=-1 reverses y, and a nonidentity odd-order point cannot have y=0. Thus a has order three and ell is 1 modulo 3.

The three x-coordinates in each orbit are the roots of a cubic with zero quadratic coefficient. Their sum is zero. Summing over H gives 2*S1=0, hence S1=0. **Nonzero S1 therefore excludes a false complete y-function match.** The converse is not asserted.



F103 CM control: order-three automorphism, degree one; all three points share y

Executed scope: 299 accepted type-II kernels from the recovered-interface reconstruction at 15 levels, plus 112 corrected CCR kernels at 3,5,7,11,13,17,19. These are 411 distinct kernels at 22 represented levels. Every full match also satisfies the independently recorded Frobenius trace relation.

Separate canonical corollary. Atkin's March 3 formula gives:

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S1 = (ell*gcd(12,ell-1)/2) * (E6/E4)
     * (j*Phi_j)/(F*Phi_F)
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The previous sealed derivative guards establish nonzero F, Phi_F and Phi_j. Cohort E4=1, E6=-b/2 and j are nonzero, as is the small integer normalizer. Thus, when this canonical formula supplies a valid cyclic kernel, its S1 is nonzero on all 886 rational canonical parameters. These are conditional mathematical deductions, not 886 new native matching calls.

Together with the 299 directly tested type-II parameters, the conditional statement spans the 1,185 parameters in the available 61-level equation set. The 112 low-level CCR calls are not added again to that parameter count. SOURCE-COROLLARY.json binds the prior proof hashes and modern-table assumption.

Evidence, controls and historical limits

Identity and sampling. All 21 models are y^2=x^3-3x+b over Fp. Exact fields, b values, orders, seeds, ordinals and UIDs are in rows.json and frozen before execution. The 11 omissions are reconstructed positions, not documented historical rejections. The input kernel census is source-pinned and complete within its stated two antecedent reconstructions.

Local catalog controls. Complete-manifest av_fq classes 1.103.au and 1.101.d have q=103/101, stored Weil polynomials 1-20T+103T^2 / 1+3T+101T^2 and p-rank one. Extraction ended 2026-10-19T16:45:49.243703+00:00 from public.av_fq_isog at devmirror.lmfdb.xyz. These are isogeny classes, not individual models. The exact small models are independently counted as 84 and 105; no cryptographic-curve catalog crosswalk is claimed.

Integrity. All 830 planned implementation calls completed; exact match sets agree between GP affine addition and FLINT formal derivative jets. Positive controls expose four false y-match scalars, including one wrong first trace. The ordinary A/A matches exactly. The formal export check passes. GP's retained stack-resize warning is nonfatal; no native error or timeout occurred.

Whole-process elapsed diagnostics are 135.0882 seconds for GP and 245.1762 for the independent verifier, on a shared desktop. These processes perform different work; no speed ratio or historical runtime is inferred. A further archive replay is validation, not a new independent sample.

Retained preparation and packaging. Five pre-run schema findings were corrected before the frozen commit. A publication-helper source review found symlink resolution could point back into sealed evidence; evidence-replay.tar.zst corrects only that helper, preserving the first archive. A presentation-only CSV converts CRLF to LF with exact row-equivalence binding. Raw scientific data and frozen source remain unchanged.

Historical boundary. The matching implementations are new reconstructions of Atkin's prose, not recovered eigenvalue code. Original ZL, CHARLA, period tables and the NIST generation caller remain unestablished. The result does not cover a single-point fingerprint, an improper kernel factor or arbitrary historical implementation faults. The original selector remains open; this round promotes no official hypothesis status.

Both canonical object-store uploads returned HeadObject403; exact archives remain in Git. The original correspondence stays private; hashes and URLs are public. Full text of the later Dewaghe paper was not acquired after AMS403, CiteSeer429 and an unreachable mirror; no algorithmic result relies on it. Existing CCR/interface/coordinate diagrams remain unchanged; the new orbit is a point diagram on one control curve.

Atkin, March 5, 1993: final calculations (retrieved; exact source hashes retained)

Atkin, March 3, 1993: normalization (retrieved; exact source hashes retained)

Evidence: AUDIT.json; EXPORT-CHECK.json; CASE-BINDINGS.json; SOURCE-COROLLARY.json; BUNDLE-REPLAY.json; archive replay receipts. Cohort SHA256: 93ac4774b993a994a17d9327df92e0f2b287e66755cd933fdc683015fec1d545.